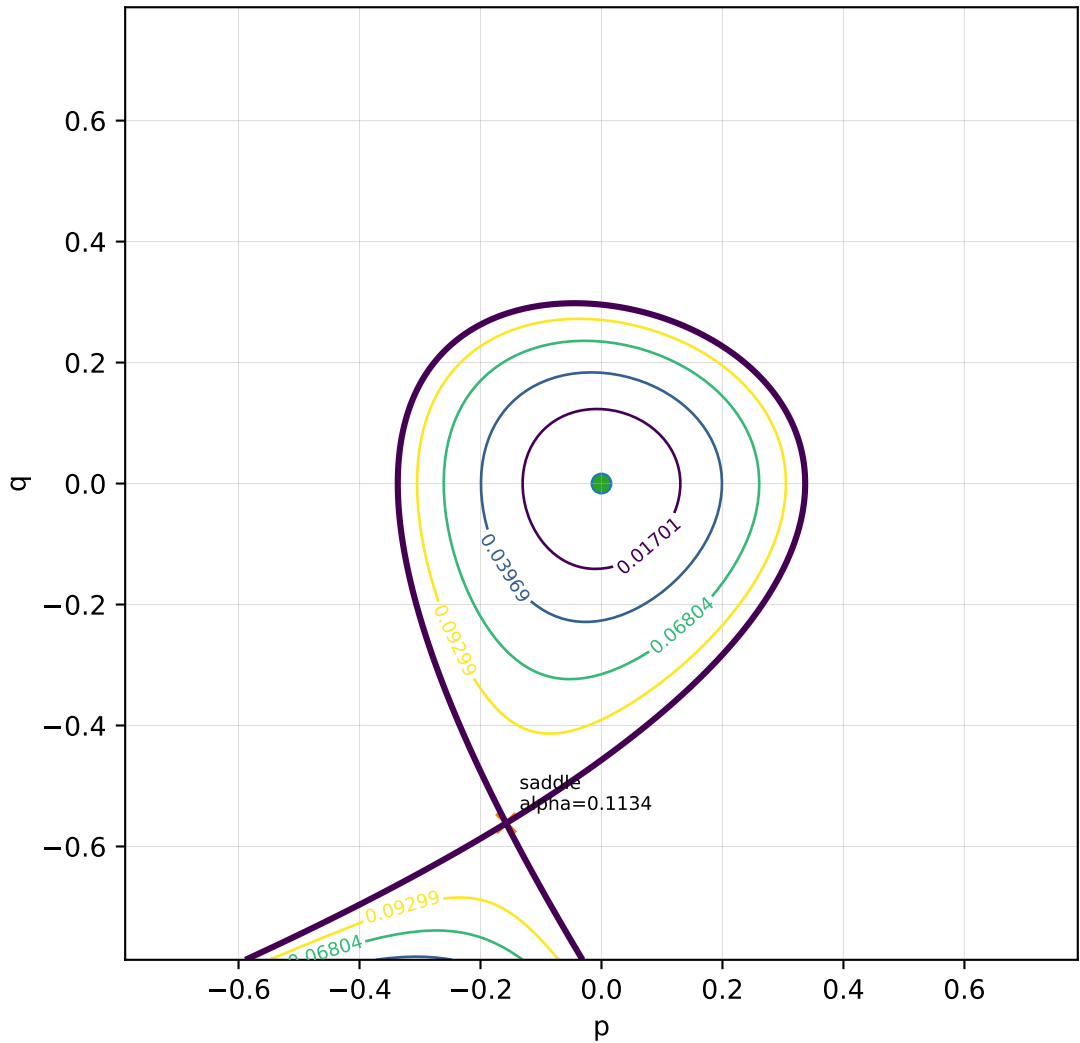


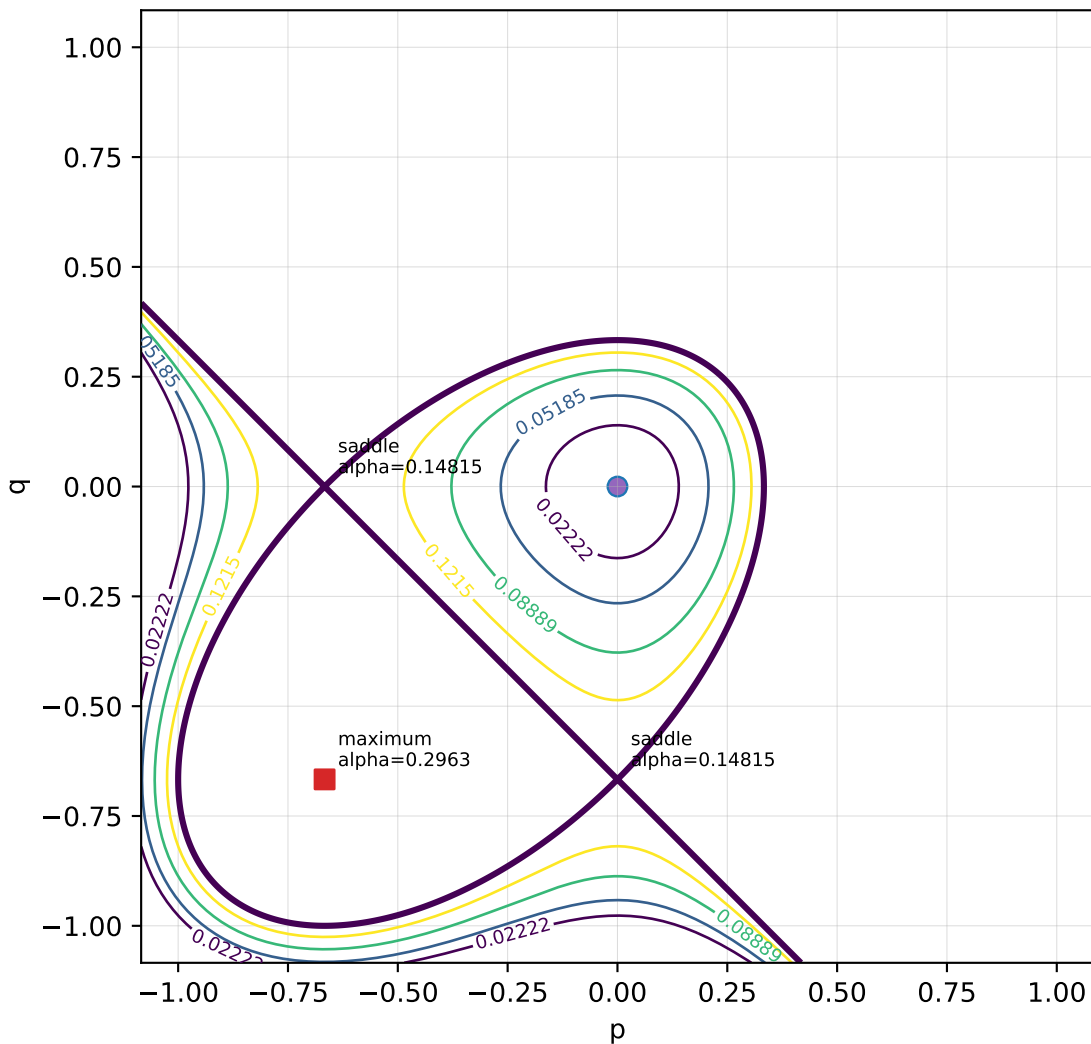


02. I1 I1 I2 IV*

harmonic plus cubic | bounding saddle energy $\alpha^* \sim 0.1481481481$

$$2H = p^{**3} + p^{**2} + q^{**3} + q^{**2}$$

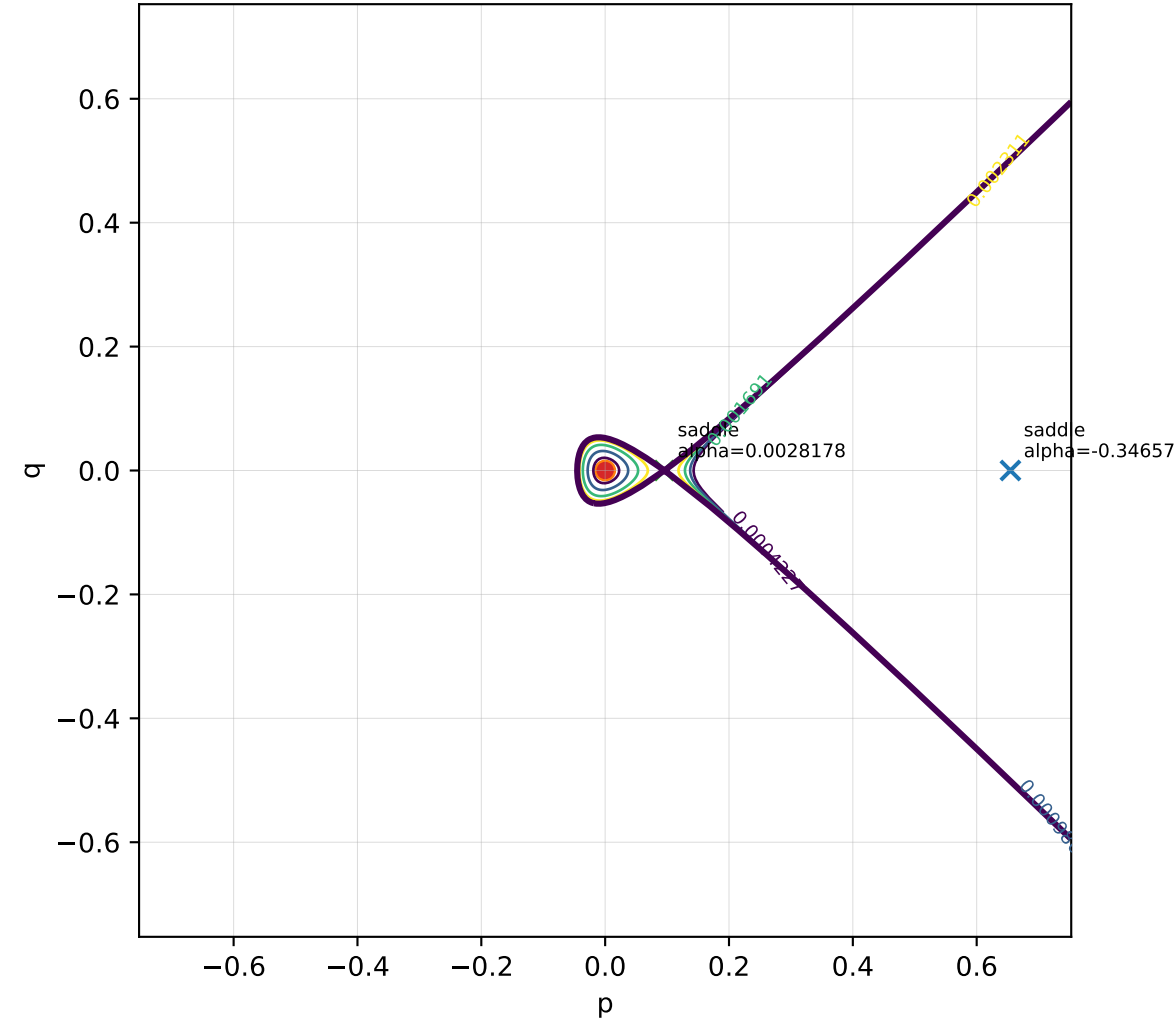
Real oscillation disk and critical level



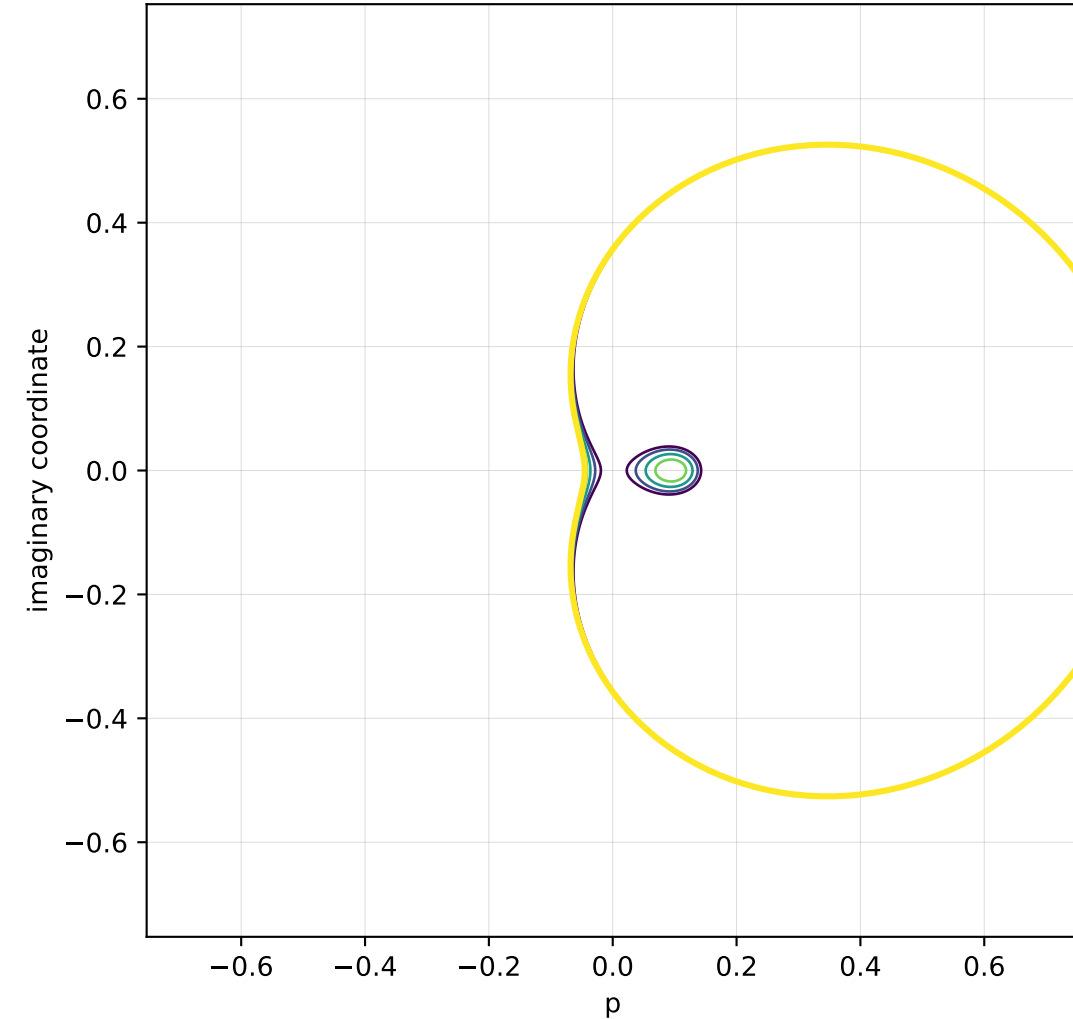


04. I1 I1 I1 I3*
 two node structured quartic | bounding saddle energy $\alpha^* \sim 0.002817810742$
 $2H = 8p^4 - 8p^3 - 16p^2q^2 + p^2 + 8pq^2 + 8q^4 + q^2$

Real oscillation disk and critical level



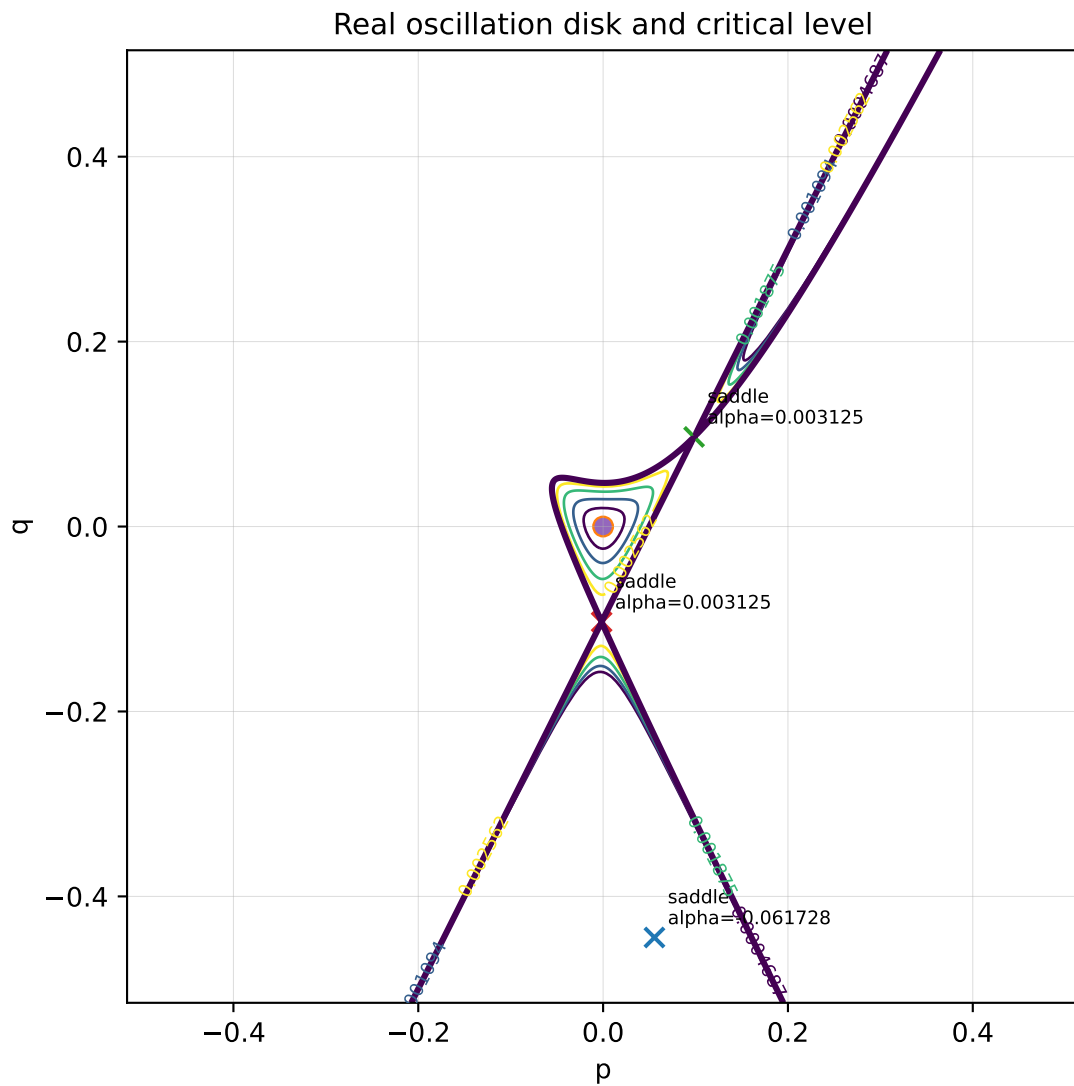
Abel-Wick real trace ($q \rightarrow iq$)



05. I1 I1 I2 I2*

two node structured quartic | bounding saddle energy $\alpha^* \sim 0.003125$

$$2H = 180p^{**4} - 4p^{**3} - 90p^{**2}q^{**2} - 32p^{**2}q + p^{**2} + p*q^{**2} + 45q^{**4}/4 + 8q^{**3} + q^{**2}$$

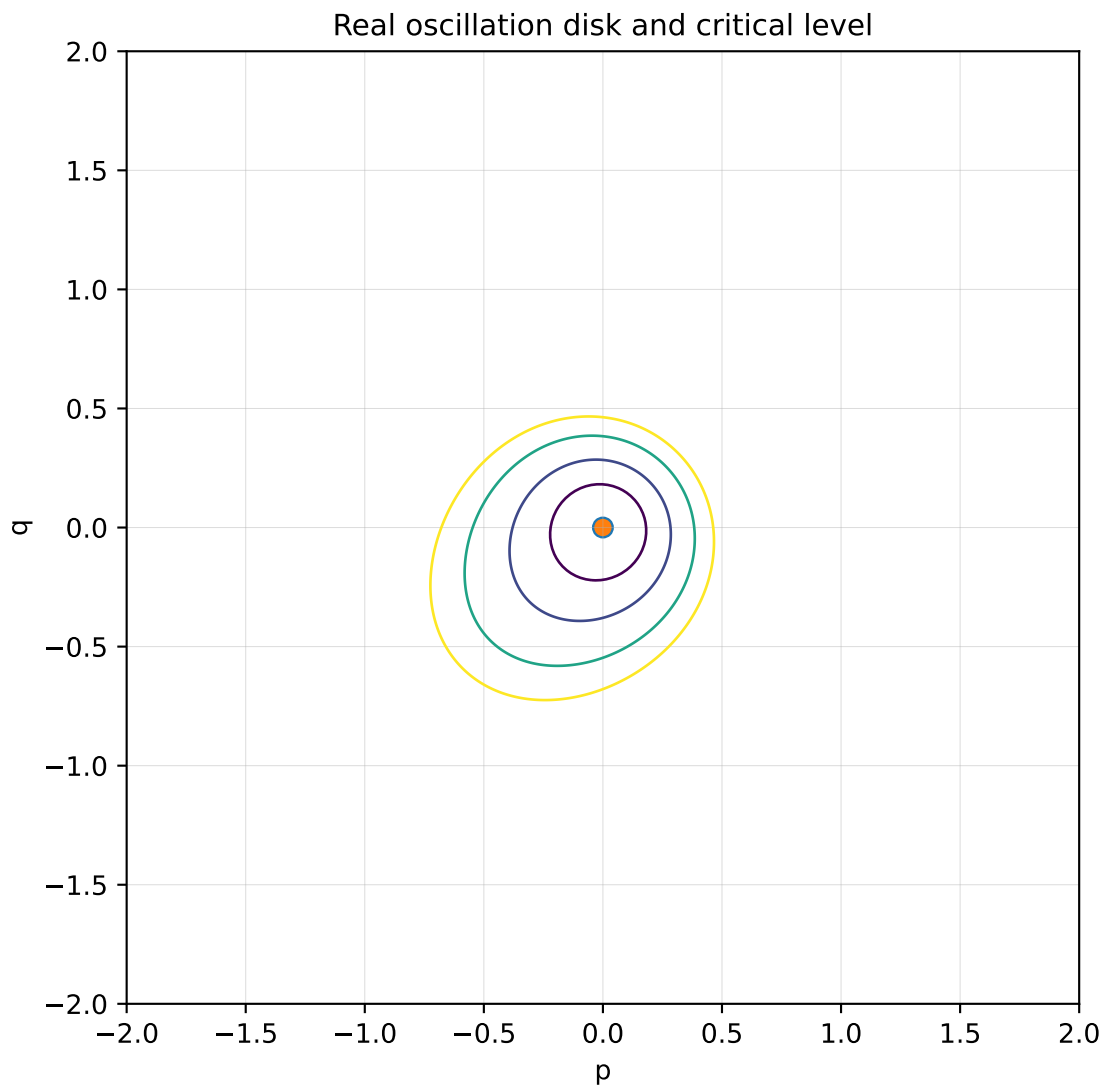


two node structured quartic | bounding saddle energy $\alpha^* \sim 0.0002931751269$

07. I1 I1 I1* I3

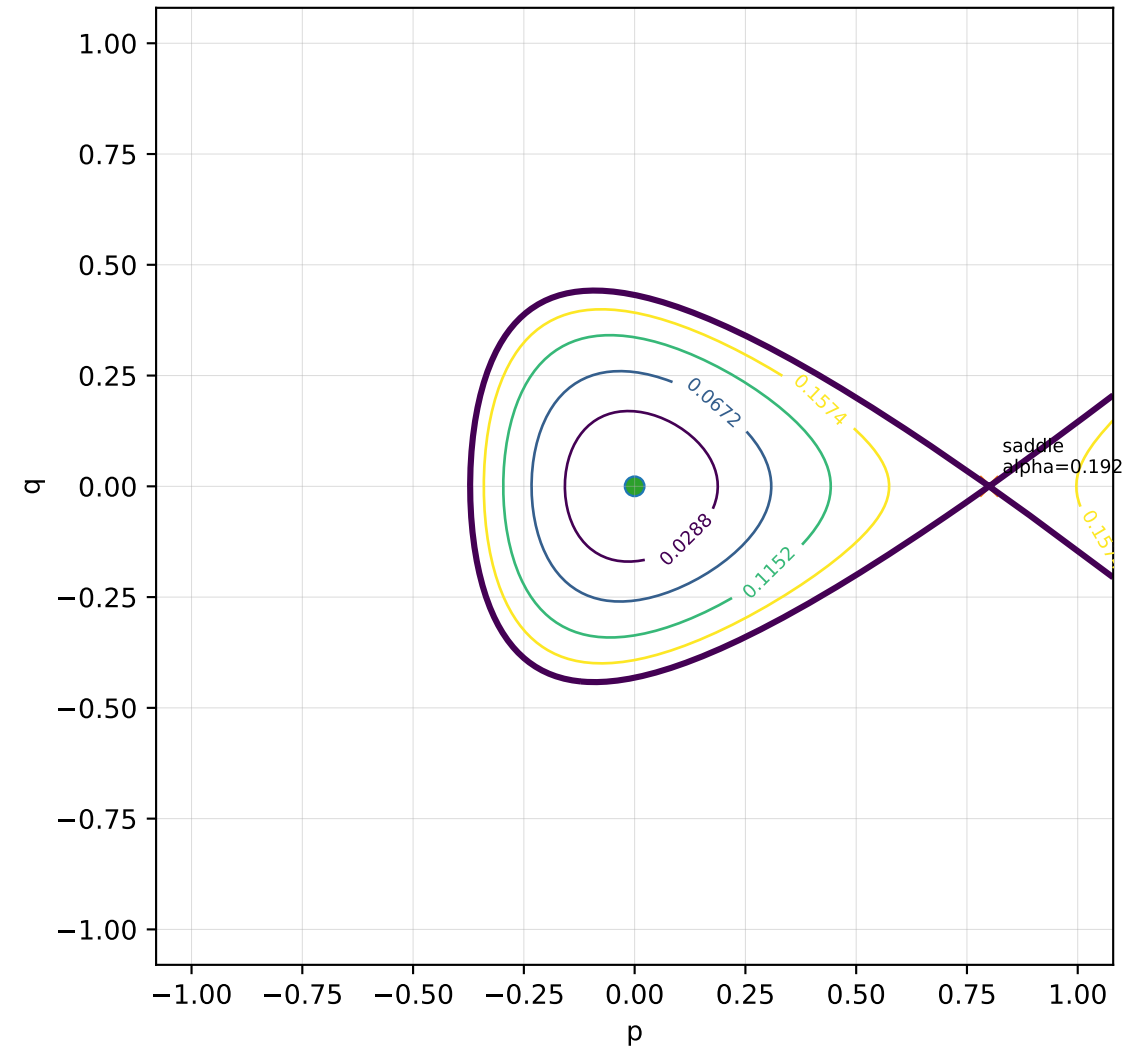
two node structured quartic | no finite positive saddle detected numerically

$$2H = p^{**4} + p^{**3} + 2*p^{**2}*q^{**2} + p^{**2}*q + p^{**2} + p*q^{**2} + q^{**4} + q^{**3} + q^{**2}$$

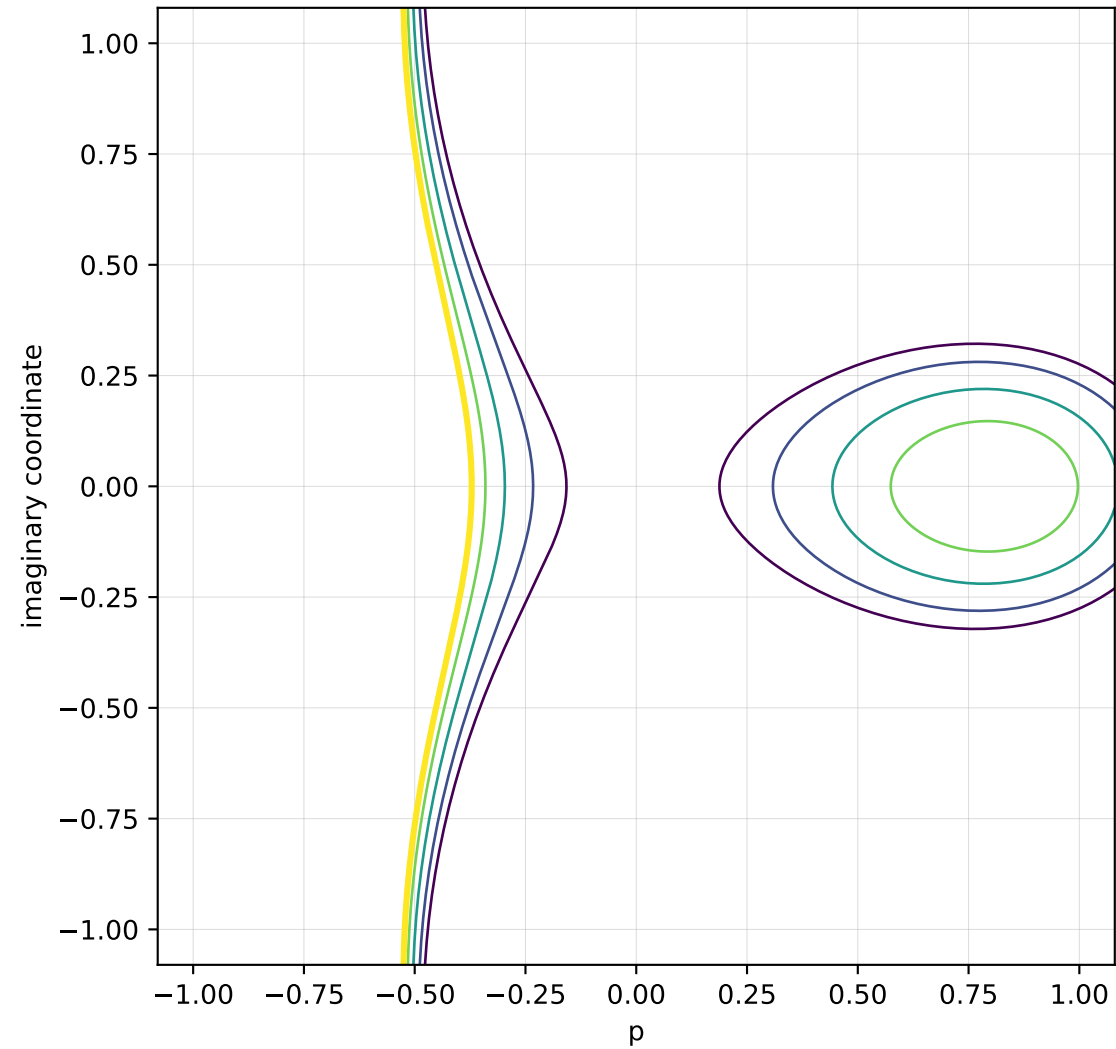


08. I1 I1 I1* III
 two node structured quartic | bounding saddle energy $\alpha^* \sim 0.192$
 $2H = 5p^{4/32} - p^3 - 5p^2q^{2/16} + p^2 + p^2q^2 + 5q^{4/32} + q^2$

Real oscillation disk and critical level

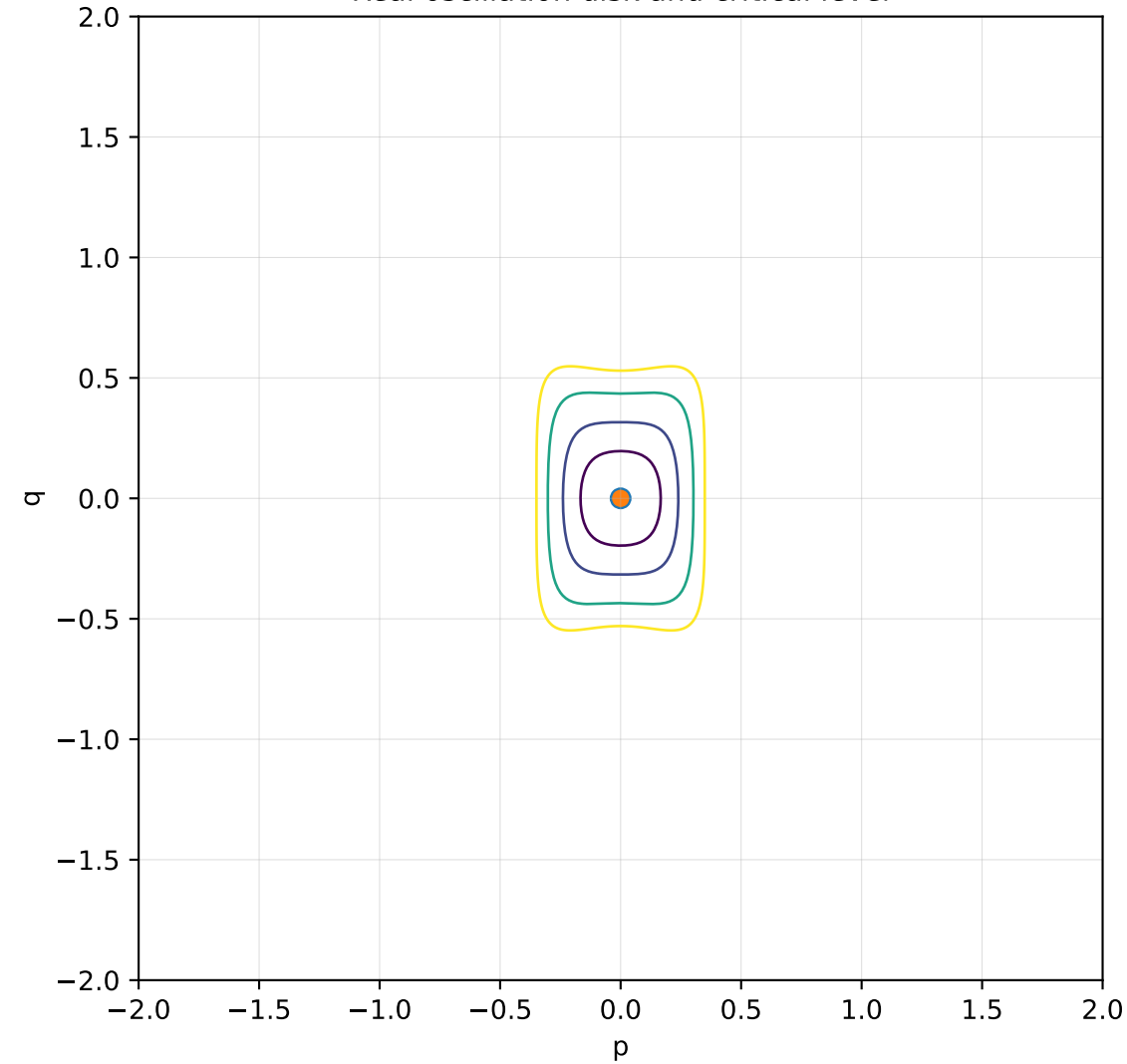


Abel-Wick real trace ($q \rightarrow iq$)

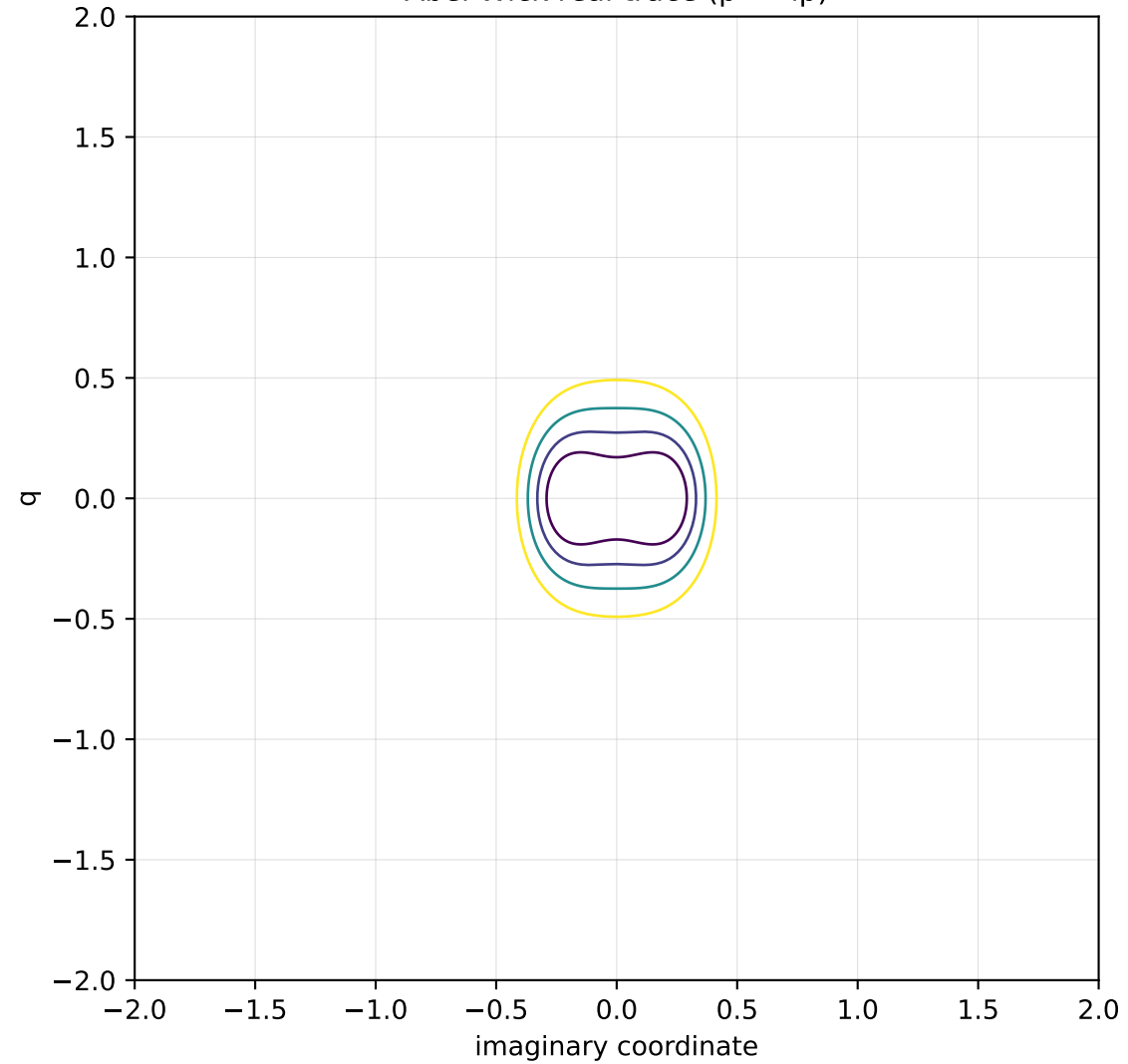


09. I1 I1* I2 I2
two node structured quartic | no finite positive saddle detected numerically
 $2H = 16p^{**4} - 8p^{**2}q^{**2} + p^{**2} + q^{**4} + q^{**2}$

Real oscillation disk and critical level

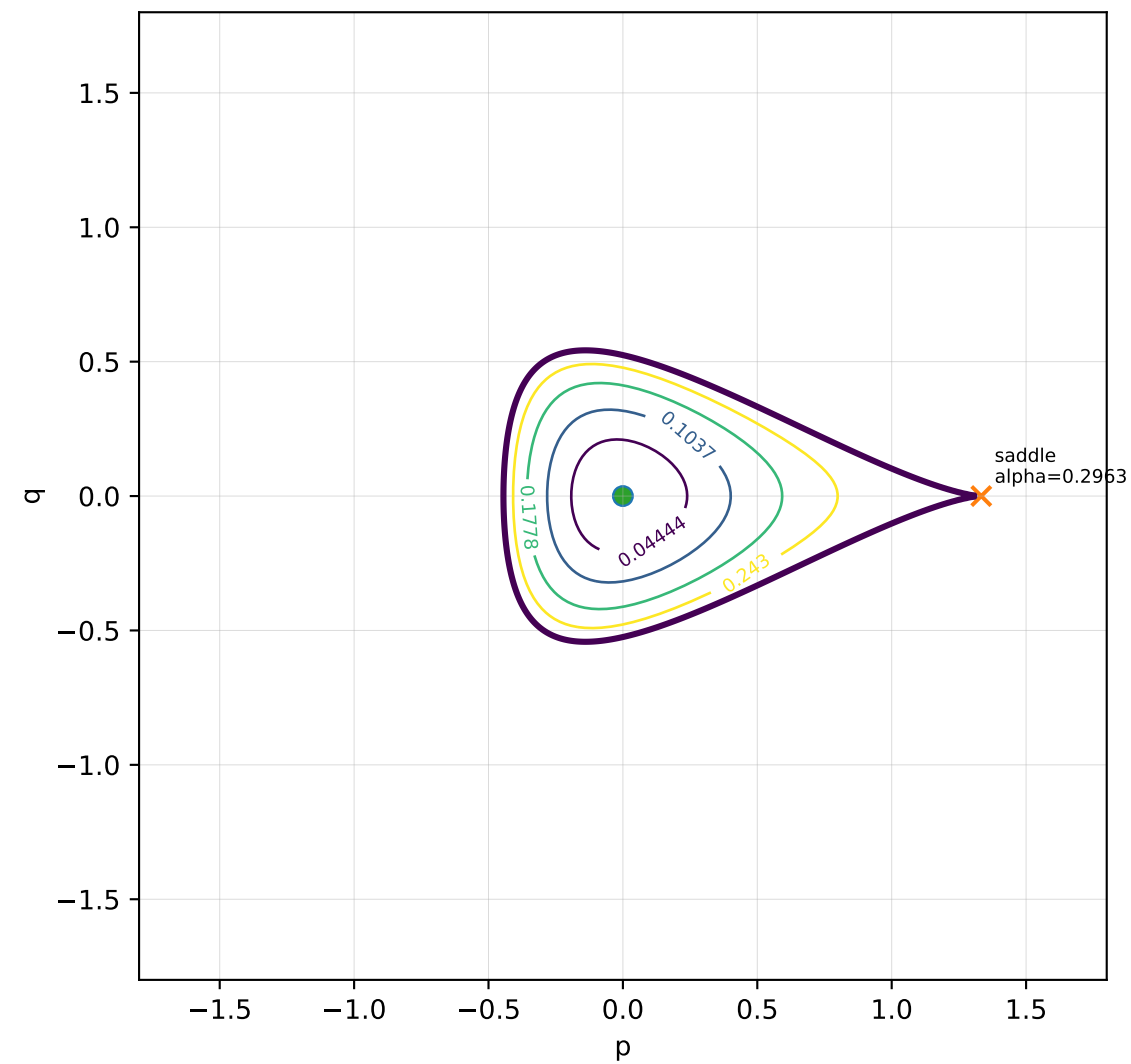


Abel-Wick real trace ($p \rightarrow ip$)

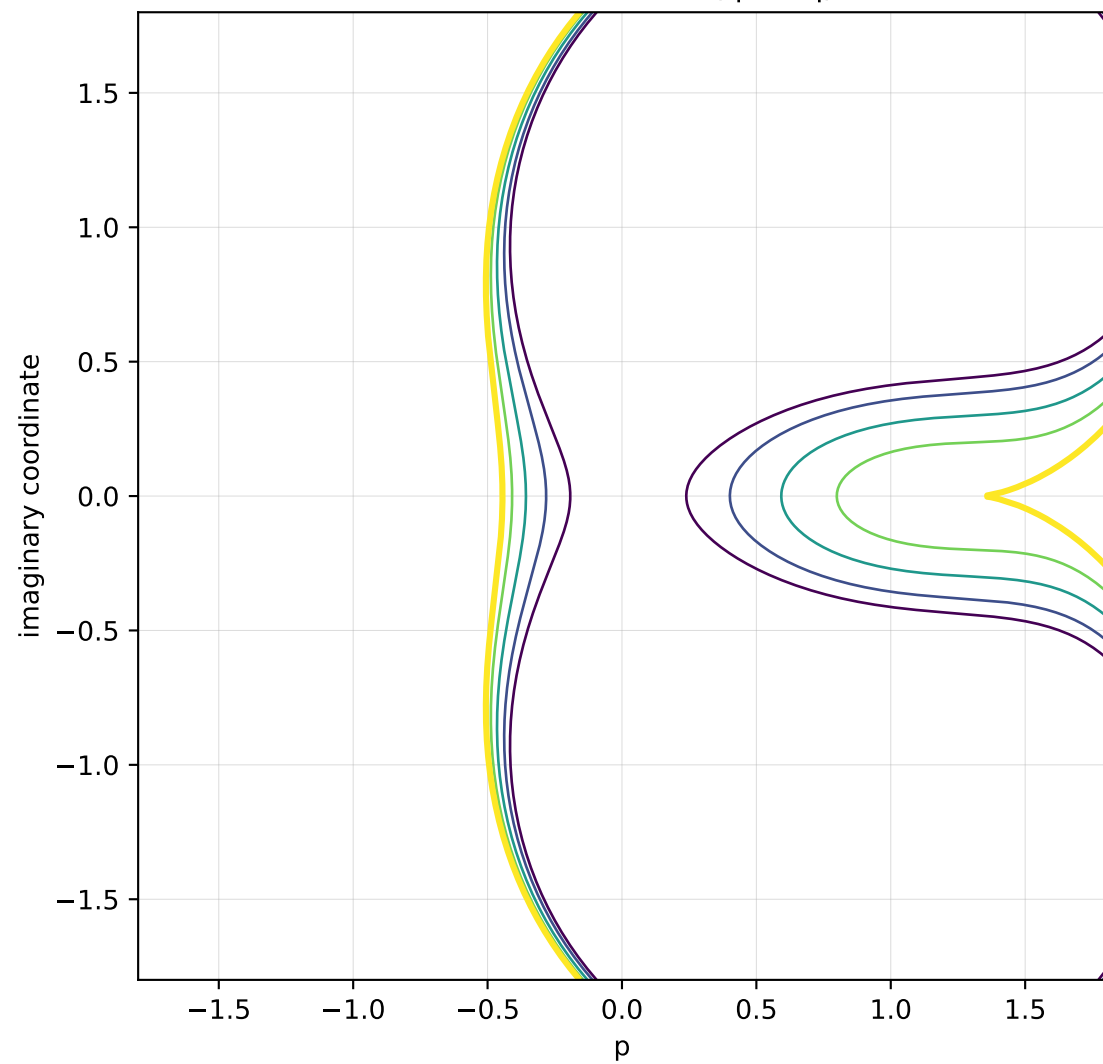


10. I1 I1* I2 II
two node structured quartic | bounding saddle energy $\alpha^* \sim 0.2962962963$
 $2H = 9p^{4/32} - p^3 - 9p^2q^{2/16} + p^2 + p^2q^2 + 9q^{4/32} + q^2$

Real oscillation disk and critical level



Abel-Wick real trace ($q \rightarrow iq$)



11. I1 I1* II II

two node structured quartic | bounding degenerate energy $\alpha^* \sim 0.5814127766$

$$2H = p^{**4}/4 - 42*\text{sqrt}(2)*p^{**3}/73 - p^{**2}*q^{**2} - 30*\text{sqrt}(2)*p^{**2}*q/73 + p^{**2} + 84*\text{sqrt}(2)*p*q^{**2}/73 + q^{**4} + 60*\text{sqrt}(2)*q^{**3}/73 + q^{**2}$$

